A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2.7 - 2.45 = 0.25 billion = 250 million years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (million years) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans **1** Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Oxygen in the atmosphere absorbed ultraviolet light **1** A dynamic equilibrium of reactions between O, O_2 and O_3 (the ozone layer) developed **1** Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) **1** Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) **1** The lower CO_2 concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion **1** Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas **1** Regulation of atmospheric composition (eg oxygen / CO_2 by photosynthesis and respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Transpiration** → water lost as vapour from the leaves **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Plants close their stomata to reduce transpiration losses **1** Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop **1** Less water for nutrient absorption / transport and cell turgidity – severe shortage dehydrates cells and inhibits biochemical reactions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms – higher risk of fungal disease – root respiration inhibited **up to 2** Drainage ditches or pipes **up to 2** Deep ploughing – avoiding compaction by machinery / livestock – adding organic matter to encourage worms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **denitrifying** → nitrate converted back to nitrogen gas • **nitrifying** → ammonium oxidised to nitrite and then nitrate	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Groundwater often contains dissolved salts – when the water evaporates from the soil the salts are left behind and accumulate **1** High soil salinity causes osmotic dehydration of roots / crop death – washing out with extra water uses more water and salinises rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **salinisation** → salt building up in the soil / soil becoming salty	2
03.4	1 mark for the point below. **1** Add crushed lime (calcium carbonate / hydroxide) – low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Size:** shows whether the population is viable / how urgent action is – sets a baseline to judge success – needed to estimate carrying capacity and sustainable off-take **2 Distribution:** shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors – identifies key areas to protect **2** Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor **2 Age structure:** many young indicates a growing population – few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous – k-selected species need breeding adults protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support • **sustainable** → meeting present needs without stopping future generations meeting theirs • **limiting factor** → the factor in shortest supply that holds the rate or population back **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	8
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition **1** Sub-sample counts in known areas scaled up to the whole reserve – mark-release-recapture (Lincoln index) for smaller species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Designation of protected areas (SSSIs, MCZs) **1** Protection of wild birds and their nests (except some pests / game birds) **1** Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts **1** Uprooting wild plants generally illegal **1** Bats must not be disturbed – roof timber treatments must be non-toxic to bats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
05.2	**1 mark each** 1 mark for each of the 4 points below. **1 Appendix I:** species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding **1 Examples:** great apes, big cats, rhinos, blue whale **1 Appendix II:** species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected **1 Examples:** Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	4
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market **1** Removing the profit removes the incentive to collect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding) – species higher up food chains need larger areas (lower density) **1** Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) **1 Shape:** the perimeter has edge conditions (a mix of neighbouring habitats) – a compact shape maximises core habitat for species needing core conditions **1** Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Varied topography:** banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) **1** Islands provide roosting / nesting sites safe from terrestrial predators (foxes) **1** Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet) – scrapes for skylarks – different water depths for waders and ducks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mobile species (birds, flying insects) colonise easily **1** Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced – keystone species especially **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community	2
07.4	1 mark for the point below. **1** Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out – avoids hazards (eg rope bridges for howler monkeys over roads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1 South America:** high loss (5.1-5.5 Mha yr^{-1}) falling by about half by 2020 **1 Africa:** loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest **1 Europe:** negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 South America:** clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export – HEP reservoirs – roads improving access – mining **up to 2 Africa:** growing rural populations clearing land for subsistence farming – fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy) – fodder cutting in dry areas **up to 2 Both:** hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted – urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** They create barriers fragmenting wildlife populations **1** Improved access makes it easier to remove timber and farm produce, so clearance spreads along them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	1 mark for the point below. **1** Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades – slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed **1** Yield mapping of different parts of a field controls fertiliser application in each area for future crops **1** This maximises yield while avoiding wastage / reducing leaching and nutrient pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive **1** They are mainly available in affluent societies – poorer farmers cannot afford the inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
09.3	1 mark for the point below. **1** Provides an income that allows producers to afford basic rights such as water, education, health care and food **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrates are leached into water bodies used as sources of potable water **1** High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2
09.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Anaerobic digestion by microbes in rice padi fields **1** Livestock intestines (ruminant digestion) **1** Anaerobic decay of manure / slurry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9

Do not write outside the box

Question	Marking guidance	Marks
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	
11.2	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Do not write outside the box